

voice

easier audio analysis

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1.Mental health

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- Emotional valence
 - The positive or negative quality of an emotion

2. Why another package

2. Why another package

Package	<i>Rec/play</i>	<i>Conversion</i>	<i>Photo/Video</i>	<i>Visualization</i>	<i>Features</i>	<i>Sheet music</i>	<i>Remove vocals</i>	<i>Chords/scales</i>	<i>Diarization</i>
audio	✓								
av	✓	✓	✓	✓					
EMU-SDMS	✓	✓		✓	✓				
gm	✓			✓		✓		✓	
karaoke							✓		
music	✓			✓				✓	
seewave	✓	✓		✓	✓				
signal		✓		✓	✓				
sound	✓								
tabr	✓			✓		✓			
tuneR				✓	✓				
voice		✓		✓	✓	✓		✓	✓
voiceR								✓	
wrassp					✓				
BirdNET-Analyzer	✓	✓		✓	✓				
Librosa				✓	✓			✓	
Parselmouth	✓	✓		✓	✓				
pyannote-audio	✓	✓		✓	✓				✓
SpeechBrain	✓	✓		✓	✓				✓

Some R and Python packages for handling audio

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signal		✓		✓	✓				
sound	✓								
tabr	✓			✓		✓			
tuneR				✓	✓				
voice		✓		✓	✓	✓		✓	✓
voiceR								✓	
wrassp					✓				
BirdNET-Analyzer	✓	✓		✓	✓				
Librosa				✓	✓			✓	
Parselmouth	✓	✓		✓	✓				
pyannote-audio	✓	✓		✓	✓				✓
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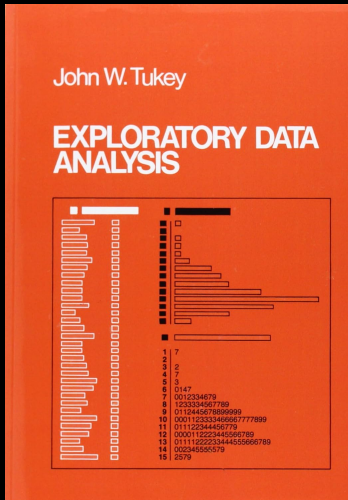
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- Allows the use of music notation and theory (ongoing work)

4. x -number summaries

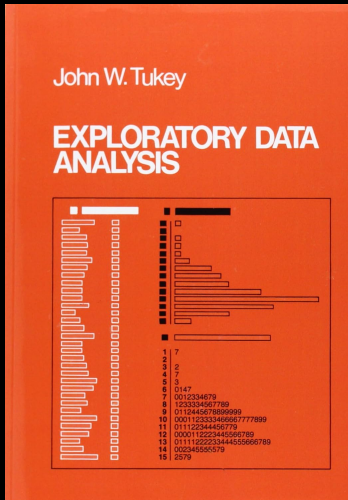
4. x-number summaries

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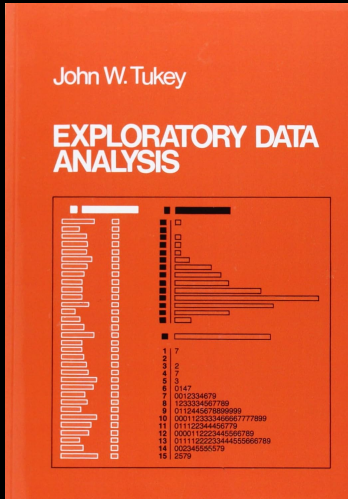


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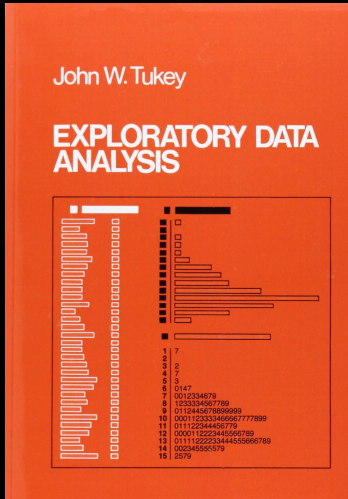


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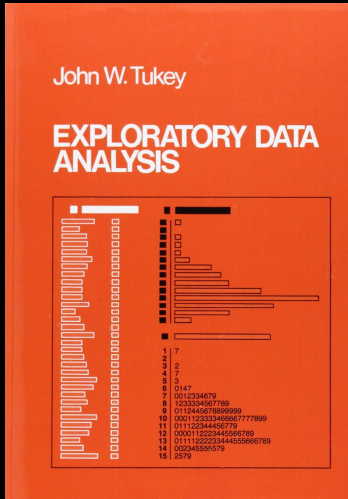
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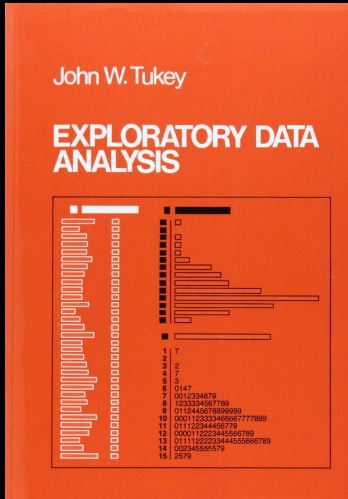
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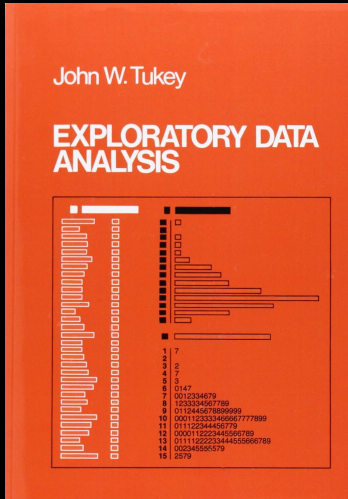
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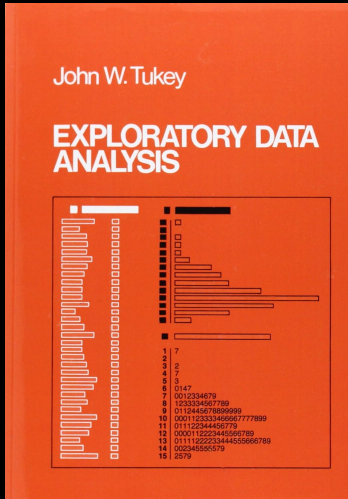


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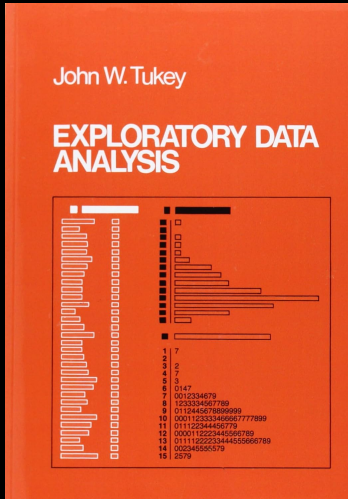
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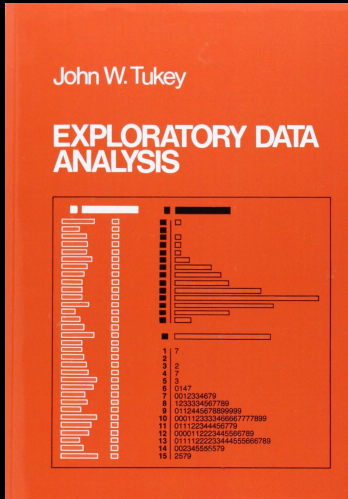
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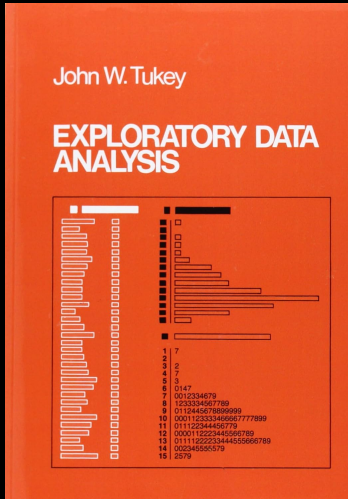
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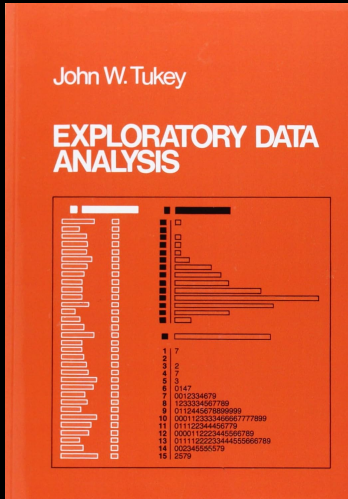
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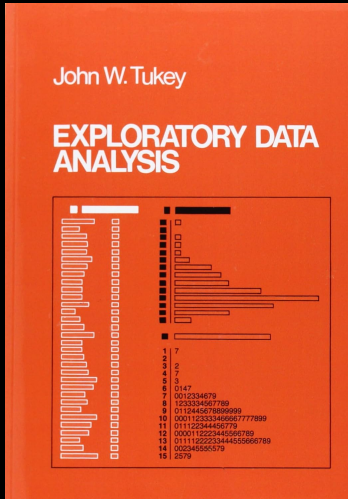
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 - 1 Mean
 - 2 Standard deviation

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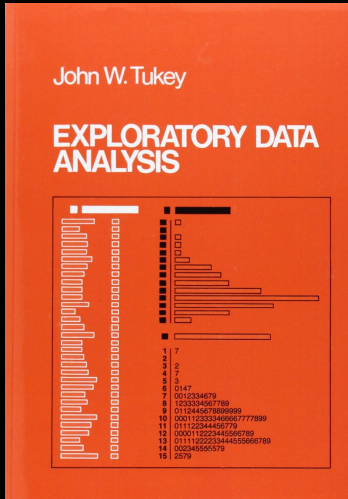
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 - 3 Coefficient of variation

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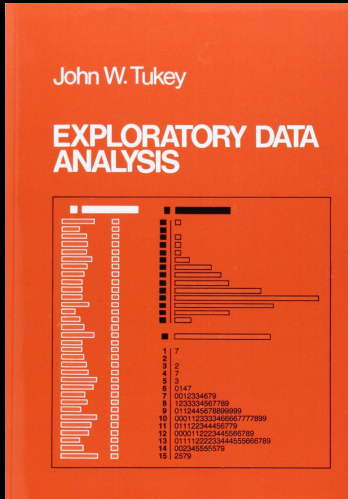
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 - ④ Median
 - ⑤ Interquartile range
 - ⑥ Median absolute deviation

5. Article 1

- voice: A Comprehensive R Package for Audio Analysis

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 - Highlights the package's performance and availability

1. Extract features

1.1 Load packages and audio files

```
# packs
library(voice)
library(tidyverse)

# get path to audio file
wavDir <- list.files(system.file('extdata', package = 'wrassp'),
                     pattern = glob2rx('*.wav'), full.names = TRUE)
```

1.2 Extract features

```
# minimal usage
M <- voice::extract_features(wavDir)
glimpse(M)
#> Rows: 1,196
#> Columns: 13
#> $ section_seq      <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16...
#> $ section_seq_file <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16...
#> $ wav_path         <chr> "/Library/Frameworks/R.framework/Versions/4.5-arm64/R...
#> $ f0               <dbl> NA, NA, NA, NA, NA, NA, NA, 115.8593, 108.9439, 107.4...
#> $ f1               <int> NA, NA, NA, NA, 185, 260, 254, 277, 261, 231, 177, 19...
#> $ f2               <int> 1854, 1886, 1749, 1888, 1962, 1973, 2026, 2037, 2130,...
#> $ f3               <int> NA, 2893, 2676, 2659, 2639, 2676, 2993, 2932, 3016, 2...
#> $ f4               <int> 3113, 3708, 3509, 3658, 3248, 3239, 3830, 3479, 3561,...
#> $ f5               <int> 4191, 4678, 4502, 4331, 3653, 3836, 4602, 4585, 4720,...
#> $ f6               <int> 5226, 5659, 5035, 5177, 5208, 5146, 5233, 5390, 5366,...
#> $ f7               <int> 6077, 6725, 6526, 6518, 6493, 6567, 6603, 6532, 6510,...
#> $ f8               <int> 6675, NA, NA, NA, 7681, 7751, 7803, NA, 7835, 7614, 7...
#> $ gain             <dbl> 21.63347, 22.76034, 28.52825, 29.67069, 36.25124, 43...
```

2. Tag

```
# creating Extended synthetic data
E <- dplyr::tibble(subject_id = c(1,1,1,2,2,2,3,3,3), wav_path = wavDir)
E
#> # A tibble: 9 × 2
#>   subject_id wav_path
#>   <dbl> <chr>
#> 1         1 /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/libra...
#> 2         1 /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/libra...
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#> 7         3 /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/libra...
#> 8         3 /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/libra...
#> 9         3 /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/libra...
```

```

# minimal usage
voice::tag(E)
#> # A tibble: 9 × 7
#>   wav_path    f0_tag_mean f0_tag_sd f0_tag_vc f0_tag_median f0_tag_iqr f0_tag_mad
#>   <chr>          <dbl>    <dbl>    <dbl>         <dbl>    <dbl>    <dbl>
#> 1 /Library/...    85.4     17.6    0.206         76.1     29.4     7.53
#> 2 /Library/...    85.4     15.6    0.183         80.1     27.8     14.4
#> 3 /Library/...    84.6     13.0    0.154         78.8     23.9     14.0
#> 4 /Library/...    84.8     14.5    0.171         79.1     28.1     11.9
#> 5 /Library/...    86.0     14.7    0.170         78.7     30.0     11.0
#> 6 /Library/...    82.9     15.6    0.188         74.8     23.8      4.78
#> 7 /Library/...    78.2     16.2    0.207         73.5     13.4      6.82
#> 8 /Library/...    84.5     14.5    0.172         78.1     17.8      8.95
#> 9 /Library/...    81.0     12.2    0.151         75.9     23.1      9.14

# canonical data
voice::tag(E, groupBy = 'subject_id')
#> # A tibble: 3 × 7
#>   subject_id f0_tag_mean f0_tag_sd f0_tag_vc f0_tag_median f0_tag_iqr f0_tag_mad
#>   <dbl>          <dbl>    <dbl>    <dbl>         <dbl>    <dbl>    <dbl>
#> 1     1         85.1     15.3    0.180         78.3     26.8     11.9
#> 2     2         84.6     14.9    0.176         76.4     28.3      7.97
#> 3     3         81.0     14.6    0.180         75.6     21.6      8.68

```

5. Diarize

```
# download
url0 <- 'https://github.com/filipezabala/voiceAudios/raw/main/wav/sherlock0.wav'
wavDir <- normalizePath(tempdir())
download.file(url0, paste0(wavDir, '/sherlock0.wav'), mode = 'wb')
```

Diarization can be performed to detect speaker segments (i.e., 'who spoke when').

```
# diarize
voice::diarize(fromWav = wavDir, toRttm = wavDir, token = 'YOUR_TOKEN')
#> Time difference of 27.49933 secs
```

The `voice::diarize()` function creates Rich Transcription Time Marked (RTTM)¹ files, space-delimited text files containing one turn per line defined by NIST - National Institute of Standards and Technology. The RTTM files can be read using `voice::read_rttm()`.

```
# read_rttm
(rttm <- voice::read_rttm(wavDir))
#> $doremi.rttm
#>      type  file chnl  tbegin  tduration ortho stype      name conf slat
#> 1 SPEAKER doremi    1 0.031 5.805  <NA>  <NA>  SPEAKER_00 <NA> <NA>
#>
#> $sherlock0.rttm
#>      type      file chnl  tbegin  tduration ortho stype      name conf slat
#> 1 SPEAKER sherlock0    1 0.908 5.231  <NA>  <NA>  SPEAKER_00 <NA> <NA>
#> 2 SPEAKER sherlock0    1 6.933 6.463  <NA>  <NA>  SPEAKER_00 <NA> <NA>
#> 3 SPEAKER sherlock0    1 13.565 8.674  <NA>  <NA>  SPEAKER_00 <NA> <NA>
```

Finally, the audio waves can be automatically segmented.

```
# split audio wave
voice::splitw(fromWav = wavDir, fromRttm = wavDir, to = wavDir)
#> TOTAL TIME 0.262 SECONDS
dir(wavDir, pattern = '.[Ww][Aa][Vv]$')
#> [1] "doremi_split_1.wav"      "doremi.wav"              "sherlock0_split_1.wav"
#> [4] "sherlock0_split_2.wav"  "sherlock0_split_3.wav"  "sherlock0.wav"
```

6. Article 2

- Predicting sex and emotional valence automatically from voice

- Predicting sex and emotional valence automatically from voice
 - Submitted to Journal of Biomedical Informatics (JBI-Elsevier)

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 - Models achieve accuracy statistically superior to the No Information Rate, the largest proportion of the observed classes
 - RF and SVM consistently demonstrate strong performance across all evaluated variables and quality measures
 - Emotional valence classification is feasible but may require hyperparameter optimization

	RAVDESS	CREMA-D
Rows/audios	1440	7442
Speakers	24	91
Language	English	English
Sex (M/F)		
#	720/720	3930/3512
%	50/50	52.8/47.2
Emotional Valence (-/+)		
#	672/768	5084/2358
%	53.3/46.7	68.3/31.7

Datasets summary

- Models

- Models
 - BL - Binary Logistic

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 - RF - Random Forests

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- 9 quality measures evaluated

- Models
 - BL - Binary Logistic
 - RF - Random Forests
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 - BART - Bayesian Additive Regression Trees
- 70%-30% train-test split
- 1,000 runs
- 9 quality measures evaluated
- Results presented with Tukey's 5-number summary

NIR: No Information Rate

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- The largest proportion of the observed classes

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- Smaller AccuracyPValue, stronger the hypothesis
Accuracy > NIR

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NIR: No Information Rate

- The largest proportion of the observed classes
- Smaller AccuracyPValue, stronger the hypothesis
Accuracy > NIR

Under $H_0 : \text{Accuracy} \leq \text{NIR}$ vs $H_1 : \text{Accuracy} > \text{NIR}$,

$$\begin{aligned}\text{AccuracyPValue} &= P(X \geq TP + TN | \text{Accuracy} = \text{NIR}) \\ &= \sum_{i=TP+TN}^n \binom{n}{i} \text{NIR}^i (1 - \text{NIR})^{n-i}\end{aligned}$$

	Accuracy					Sensitivity					Specificity				
	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%
RAVDESS															
BART	0.127	0.776	0.842	0.927	0.998	0.002	0.757	0.971	1.000	1.000	0.007	0.738	0.946	1.000	1.000
RF	0.100	0.806	0.885	0.933	0.998	0.125	0.883	0.967	0.992	1.000	0.100	0.760	0.908	0.983	1.000
SVM	0.056	0.841	0.912	0.950	1.000	0.056	0.897	0.983	1.000	1.000	0.106	0.820	0.938	0.994	1.000
BL	0.360	0.729	0.792	0.842	0.965	0.274	0.751	0.871	0.953	1.000	0.319	0.690	0.812	0.906	1.000
CREMA-D															
BART	0.284	0.857	0.907	0.929	0.978	0.129	0.859	0.940	0.978	1.000	0.179	0.859	0.931	0.974	1.000
RF	0.609	0.888	0.910	0.928	0.969	0.554	0.909	0.947	0.972	1.000	0.479	0.844	0.903	0.944	0.998
SVM	0.654	0.913	0.934	0.948	0.986	0.578	0.920	0.955	0.980	1.000	0.548	0.897	0.940	0.972	1.000
BL	0.630	0.881	0.905	0.922	0.974	0.549	0.882	0.926	0.957	0.999	0.555	0.864	0.913	0.948	1.000
	Pos Pred Value					Neg Pred Value					F1				
	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%
RAVDESS															
BART	0.126	0.698	0.947	1.000	1.000	0.125	0.710	0.972	1.000	1.000	0.005	0.771	0.842	0.930	0.998
RF	0.155	0.729	0.915	0.988	1.000	0.168	0.835	0.965	0.992	1.000	0.222	0.804	0.887	0.940	0.998
SVM	0.169	0.779	0.938	0.996	1.000	0.157	0.853	0.982	1.000	1.000	0.107	0.832	0.916	0.953	1.000
BL	0.173	0.653	0.822	0.922	1.000	0.160	0.699	0.868	0.958	1.000	0.296	0.735	0.795	0.848	0.965
CREMA-D															
BART	0.288	0.861	0.937	0.978	1.000	0.200	0.837	0.934	0.978	1.000	0.229	0.864	0.910	0.932	0.979
RF	0.389	0.851	0.915	0.957	0.999	0.327	0.883	0.941	0.971	1.000	0.560	0.894	0.917	0.933	0.971
SVM	0.423	0.894	0.947	0.978	1.000	0.340	0.898	0.952	0.979	1.000	0.595	0.918	0.937	0.951	0.987
BL	0.439	0.865	0.922	0.959	1.000	0.325	0.855	0.919	0.957	0.999	0.609	0.887	0.910	0.926	0.976
	Detection Rate					Kappa					AccuracyPValue				
	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%
RAVDESS															
BART	0.002	0.375	0.431	0.481	0.554	0.001	0.572	0.689	0.854	0.996	0.000	0.000	0.000	0.000	1.000
RF	0.000	0.373	0.475	0.498	0.604	0.000	0.620	0.760	0.863	0.996	0.000	0.000	0.000	0.000	1.000
SVM	0.000	0.375	0.481	0.498	0.604	0.000	0.685	0.821	0.897	1.000	0.000	0.000	0.000	0.000	1.000
BL	0.123	0.360	0.410	0.448	0.565	0.076	0.473	0.591	0.684	0.929	0.000	0.000	0.000	0.000	1.000
CREMA-D															
BART	0.106	0.450	0.476	0.492	0.528	0.050	0.719	0.813	0.857	0.956	0.000	0.000	0.000	0.000	1.000
RF	0.249	0.454	0.495	0.526	0.590	0.306	0.775	0.818	0.854	0.937	0.000	0.000	0.000	0.000	1.000
SVM	0.249	0.458	0.498	0.529	0.603	0.329	0.823	0.868	0.894	0.971	0.000	0.000	0.000	0.000	1.000
BL	0.248	0.448	0.482	0.511	0.567	0.303	0.761	0.810	0.843	0.949	0.000	0.000	0.000	0.000	1.000

Summary of SEX classification performance over 1,000 simulations

	Accuracy					Sensitivity					Specificity				
	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%
RAVDESS															
BART	0.606	0.698	0.717	0.735	0.804	0.625	0.707	0.727	0.746	0.828	0.536	0.674	0.705	0.737	0.826
RF	0.613	0.698	0.719	0.738	0.796	0.684	0.766	0.789	0.812	0.895	0.442	0.603	0.638	0.674	0.795
SVM	0.635	0.710	0.731	0.748	0.802	0.711	0.789	0.812	0.832	0.898	0.469	0.598	0.638	0.674	0.795
BL	0.487	0.575	0.600	0.621	0.704	0.449	0.574	0.602	0.625	0.703	0.455	0.562	0.598	0.625	0.728
CREMA-D															
BART	0.633	0.669	0.677	0.685	0.716	0.573	0.625	0.636	0.646	0.684	0.645	0.751	0.766	0.781	0.835
RF	0.704	0.725	0.731	0.736	0.764	0.902	0.935	0.944	0.952	0.970	0.181	0.255	0.275	0.294	0.373
SVM	0.718	0.740	0.746	0.753	0.775	0.863	0.900	0.912	0.920	0.952	0.309	0.376	0.393	0.409	0.492
BL	0.535	0.584	0.601	0.645	0.716	0.495	0.581	0.607	0.661	0.790	0.461	0.577	0.604	0.627	0.692
	Pos Pred Value					Neg Pred Value					F1				
	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%
RAVDESS															
BART	0.622	0.717	0.739	0.760	0.834	0.583	0.673	0.693	0.712	0.798	0.635	0.714	0.732	0.749	0.819
RF	0.623	0.693	0.714	0.733	0.815	0.596	0.701	0.724	0.747	0.846	0.654	0.732	0.749	0.765	0.818
SVM	0.635	0.698	0.720	0.740	0.817	0.635	0.724	0.747	0.768	0.853	0.685	0.746	0.762	0.776	0.825
BL	0.520	0.605	0.629	0.653	0.742	0.452	0.542	0.567	0.590	0.668	0.484	0.592	0.615	0.637	0.711
CREMA-D															
BART	0.792	0.846	0.854	0.862	0.891	0.446	0.486	0.494	0.503	0.539	0.682	0.721	0.729	0.736	0.766
RF	0.715	0.732	0.736	0.741	0.765	0.585	0.670	0.696	0.721	0.789	0.810	0.824	0.827	0.830	0.846
SVM	0.742	0.759	0.763	0.768	0.792	0.593	0.654	0.673	0.695	0.761	0.811	0.826	0.831	0.835	0.850
BL	0.709	0.751	0.766	0.786	0.834	0.353	0.394	0.413	0.456	0.544	0.592	0.657	0.675	0.718	0.777
	Detection Rate					Kappa					AccuracyPValue				
	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%	0%	25%	50%	75%	100%
RAVDESS															
BART	0.333	0.377	0.388	0.398	0.442	0.205	0.394	0.432	0.468	0.606	0.000	0.000	0.000	0.000	0.001
RF	0.365	0.408	0.421	0.433	0.477	0.216	0.389	0.430	0.469	0.587	0.000	0.000	0.000	0.000	0.000
SVM	0.379	0.421	0.433	0.444	0.479	0.259	0.412	0.454	0.490	0.602	0.000	0.000	0.000	0.000	0.000
BL	0.240	0.306	0.321	0.333	0.375	-0.028	0.147	0.198	0.239	0.409	1.000	1.000	1.000	1.000	1.000
CREMA-D															
BART	0.392	0.426	0.434	0.441	0.465	0.244	0.336	0.350	0.364	0.419	0.000	0.403	0.714	0.922	1.000
RF	0.616	0.638	0.644	0.650	0.662	0.260	0.277	0.362	0.377	0.369	0.000	0.000	0.000	0.000	0.013
SVM	0.591	0.615	0.622	0.628	0.648	0.263	0.325	0.341	0.358	0.419	0.000	0.000	0.000	0.000	0.000
BL	0.338	0.397	0.414	0.452	0.541	0.064	0.148	0.181	0.249	0.391	1.000	1.000	1.000	1.000	1.000

Summary of EMOTIONAL VALENCE classif. performance over 1,000 simulations


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Open-access code and data used in Zabala & Salum (2025) - Predicting sex and emotion valence automatically from voice

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<https://osf.io/ahrbs/files/osfstorage>

The screenshot displays the OSFHOME web interface. At the top, there is a navigation bar with the OSFHOME logo and a dropdown arrow, followed by links for 'My Projects', 'Search', 'Support', 'Donate', and a user profile for 'Filipe Zabala'. Below this is a secondary navigation bar with tabs for 'voice_audios', 'Metadata', 'Files', 'Wiki' (which is highlighted), 'Components', 'Analytics', 'Registrations', 'Contributors', 'Add-ons', 'Linked services', and 'Settings'. The main content area is titled 'Home' and includes a 'Toggle view:' button with 'View', 'Edit', and 'Compare' options. On the left, there is a sidebar with a '+ New' button and a list of 'Project Wiki Pages' including 'Home'. The main content area shows the 'View' tab selected, displaying the 'Wiki Version' as '(Current) Filipe Zabala: 2025-05-23 19:54:15+00:00 UTC'. The text describes four standardized open audio datasets converted to wav mono via `ffmpeg -i fileOriginal -ac 1 fileWavMono`. It includes a list of references for each dataset: AESDD (Vryzas et al (2018a), Vryzas et al (2018b), AESDD data, 605 audio files, 1 zip file (171.9 MB)), CREMA-D (Cao et al. (2014), CREMA-D data, 7,442 audio files, 1 zip file (1.1 GB)), and RAVDESS (Livingstone and Russo (2018), RAVDESS data, 7,356 audio files, 1 zip file (217.6 MB)).

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Project Wiki Pages

Home

View Wiki Version: (Current) Filipe Zabala: 2025-05-23 19:54:15+00:00 UTC

Four standardized open audio datasets, converted to wav mono via `ffmpeg -i fileOriginal -ac 1 fileWavMono`. If you use this repository in your work, please consider requesting to add a link to your work in the 'Description: Articles using this repository' section of this webpage.

1. AESDD
 - Vryzas et al (2018a)
 - Vryzas et al (2018b)
 - AESDD data
 - 605 audio files
 - 1 zip file (171.9 MB)
2. CREMA-D
 - Cao et al. (2014)
 - CREMA-D data
 - 7,442 audio files
 - 1 zip file (1.1 GB)
3. RAVDESS
 - Livingstone and Russo (2018)
 - RAVDESS data
 - 7,356 audio files
 - 1 zip file (217.6 MB)

<https://osf.io/9g73a/wiki/home/>

7. Current research

- Link with other areas

7. Current research

- Link with other areas
 - Linguistics (Prosody, [Cláudia Brescancini](#), PUCRS, Brazil)

- Link with other areas
 - Linguistics (Prosody, [Cláudia Brescancini](#), PUCRS, Brazil)
 - Pulmonology (Chronic cough)

- Link with other areas
 - Linguistics (Prosody, [Cláudia Brescancini](#), PUCRS, Brazil)
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 - [Waldo Mattos](#), UFCSPA/Hospital Conceição/WM Pesquisa Clínica)

- Link with other areas
 - Linguistics (Prosody, [Cláudia Brescancini](#), PUCRS, Brazil)
 - Pulmonology (Chronic cough)
 - [Waldo Mattos](#), UFCSPA/Hospital Conceição/WM Pesquisa Clínica)
 - [Cesar A. Keller](#), Mayo Clinic, USA)

- Link with other areas
 - Linguistics (Prosody, [Cláudia Brescancini](#), PUCRS, Brazil)
 - Pulmonology (Chronic cough)
 - [Waldo Mattos](#), UFCSPA/Hospital Conceição/WM Pesquisa Clínica)
 - [Cesar A. Keller](#), Mayo Clinic, USA)
- Automatically identify speakers

- Link with other areas
 - Linguistics (Prosody, [Cláudia Brescancini](#), PUCRS, Brazil)
 - Pulmonology (Chronic cough)
 - [Waldo Mattos](#), UFCSPA/Hospital Conceição/WM Pesquisa Clínica)
 - [Cesar A. Keller](#), Mayo Clinic, USA)
- Automatically identify speakers
 - Proportion of speech on audio by speaker

- Link with other areas
 - Linguistics (Prosody, [Cláudia Brescancini](#), PUCRS, Brazil)
 - Pulmonology (Chronic cough)
 - [Waldo Mattos](#), UFCSPA/Hospital Conceição/WM Pesquisa Clínica)
 - [Cesar A. Keller](#), Mayo Clinic, USA)
- Automatically identify speakers
 - Proportion of speech on audio by speaker
 - Summary statistics by speaker

- Link with other areas
 - Linguistics (Prosody, [Cláudia Brescancini](#), PUCRS, Brazil)
 - Pulmonology (Chronic cough)
 - [Waldo Mattos](#), UFCSPA/Hospital Conceição/WM Pesquisa Clínica)
 - [Cesar A. Keller](#), Mayo Clinic, USA)
- Automatically identify speakers
 - Proportion of speech on audio by speaker
 - Summary statistics by speaker
 - Probabilities by speaker

- Link with other areas
 - Linguistics (Prosody, [Cláudia Brescancini](#), PUCRS, Brazil)
 - Pulmonology (Chronic cough)
 - [Waldo Mattos](#), UFCSPA/Hospital Conceição/WM Pesquisa Clínica)
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- Build a dashboard to allow easy interaction
 - Allow user to send, link and record audios

Thank you!

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